Complex Variables and Linear Algebra

(BMAT201L)

Reference Material

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Module 3

Contour Integration

3.1 Taylor's Series

Theorem 3.1 (Taylor's Series about a Point): *Let $f(z)$ be analytic on $\mathcal{D} \subset \mathbb{C}$ and $z_0 \in \mathcal{D}$. Then the Taylor's series of f about z_0 is given by*

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots, \tag{3.1}$$

where

$$a_0 = f(z_0),\ a_n = \frac{f^{(n)}(z_0)}{n!},\ n \geq 1 \tag{3.2}$$

are called the coefficients of the Taylor's series (3.1).

✍ To find the Taylor's series of f about $z_0 \neq 0$, employ the formulas (3.1).

✍ To find the Taylor's series of f about the origin $z_0 = 0$, employ the formula

$$f(z) = \sum_{n=0}^{\infty} a_n z^n \text{ where } a_0 = f(0),\ a_n = \frac{f^{(n)}(0)}{n!},\ n \geq 1. \tag{3.3}$$

Example 3.1: Find the Taylor's series of $f(z) = e^z$ about $z_0 = 2$.

Solution: We see that $a_0 = f(2) = e^2$, and $a_n = \frac{f^{(n)}(2)}{n!} = \frac{e^2}{n!}$ for $n \geq 1$. Therefore,

$$f(z) = e^z = \sum_{n=0}^{\infty} a_n (z-2)^n = e^2 \sum_{n=0}^{\infty} \frac{(z-2)^n}{n!}. \tag{3.4}$$

The Taylor's series of $f(z) = e^z$ about $z_0 = 0$ is

$$f(z) = e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}. \tag{3.5}$$

Some Well-known Taylor's series about $z = 0$

(a) $\sinh z = \sum_{n=0}^{\infty} \left\{ \frac{z^{2n+1}}{(2n+1)!} \right\}$

(b) $\cosh z = \sum_{n=0}^{\infty} \left\{ \frac{z^{2n}}{(2n)!} \right\}$

(c) $\sin z = \sum_{n=0}^{\infty} \left\{ \frac{(-1)^n}{(2n+1)!} \right\} z^{2n+1}$

(d) $\cos z = \sum_{n=0}^{\infty} \left\{ \frac{(-1)^n}{(2n)!} \right\} z^{2n}$.

Taylor's Series of Rational Polynomial Functions about the Origin

✍ A rational polynomial function $f(z)$ is the quotient $p(z)/q(z)$ of two polynomials $p(z)$ and $q(z)$.

✍ To find the Taylor's series of a rational polynomial function f about the origin $z_0 = 0$, first reduce it into partial fractions, and then employ the binomial expansion formula to each partial fraction.

Example 3.2: Find the Taylor's series[1] of $f(z) = \frac{1}{(z-1)(z+2)}$ about $z = 0$, and determine its region of validity.

Solution: We have

$$\begin{aligned} f(z) &= \frac{1}{(z-1)(z+2)} = \frac{1}{3}\left\{\frac{1}{z-1} - \frac{1}{z+2}\right\} = \frac{1}{3}\left\{-(1-z)^{-1} - \frac{1}{2}\left(1+\frac{z}{2}\right)^{-1}\right\} \\ &= \frac{1}{3}\left\{-\sum_{n=0}^{\infty} z^n - \frac{1}{2}\sum_{n=0}^{\infty}(-1)^n\left(\frac{z}{2}\right)^n\right\}. \end{aligned}$$

The first series converges on the region $S_1 : |z| < 1$, while the second series converges on the region $|z/2| < 1$ or $S_2 : |z| < 2$. Therefore, the region of convergence of $f(z)$ is $S_1 \cap S_2 : |z| < 1$, which contains $z = 0$. Thus

$$f(z) = \frac{1}{(z-1)(z+2)} = \frac{1}{3}\sum_{n=0}^{\infty}\left\{-1 + \left(-\frac{1}{2}\right)^{n+1}\right\} z^n, \text{ valid on } |z| < 1.$$

Taylor's Series of Rational Polynomial Functions about $z \neq 0$

To find the Taylor's series of a rational polynomial function f about $z_0 \neq 0$:

✍ Substitute $z - z_0 = u$ in f to get a rational polynomial function $\psi(u)$.

✍ Reduce ψ into partial fractions, and then employ the binomial expansion formula to each partial fraction, and then simplify if needed.

Example 3.3: Find the Taylor's series of $f(z) = \frac{1}{(z-1)(z-3)}$ about $z = 4$, and determine its region of validity.

Solution: Write $u = z - 4$ or $z = u + 4$. Then we see that

$$\begin{aligned} f(z) &= \frac{1}{(u+3)(u+1)} = \frac{1}{2}\left\{\frac{1}{u+1} - \frac{1}{u+3}\right\} = \frac{1}{2}\left\{(1+u)^{-1} - \frac{1}{3}\left(1+\frac{u}{3}\right)^{-1}\right\} \\ &= \frac{1}{2}\left\{\sum_{n=0}^{\infty}(-1)^n u^n - \frac{1}{3}\sum_{n=0}^{\infty}(-1)^n\left(\frac{u}{3}\right)^n\right\}. \end{aligned}$$

The first series converges on the region $S_1 : |u| < 1$, while the second series converges on the region $|u/3| < 1$ or $S_2 : |u| < 3$. Hence, the region of convergence of $f(z)$ is $S_1 \cap S_2 : |u| < 1$, that is $|z - 4| < 1$. Thus the Taylor's series of f about $z = 4$ is

$$f(z) = \frac{1}{2}\sum_{n=0}^{\infty}\left\{1 - \frac{1}{3^{n+1}}\right\}(-1)^n(z-4)^n,$$

[1]For real α, the binomial series $(1+z)^\alpha = 1 + \alpha z + \frac{\alpha(\alpha-1)z^2}{2!} + \frac{\alpha(\alpha-1)(\alpha-2)z^3}{3!} + \cdots$ converges on $|z| < 1$.

which is valid over the interior of the circle with centre at 4 and radius 1.

Example 3.4: Find the Taylor's series of $f(z) = 1/z^2$ about $z = i$, and determine its region of validity.

Solution: Write $u = z - i$ or $z = u + i$. Then we see that

$$\begin{aligned} f(z) &= \frac{1}{(u+i)^2} = -1\Big/\left(1+\frac{u}{i}\right)^2 = -(1-iu)^{-2} \\ &= -\sum_{n=0}^{\infty}(n+1)(iu)^n = -\sum_{n=0}^{\infty}(n+1)(i)^n(z-i)^n. \end{aligned}$$

The series converges on the region $S : |iu| < 1$, that is on $|z - i| < 1$.

3.2 Laurent's Series

Definition 3.1 (Isolated singularity): If $f(z)$ is analytic on some deleted neighborhood $N(z_0) - \{z_0\}$ of $z_0 \in \mathbb{C}$, then z_0 is called an *isolated singularity* of f.

Laurent's Series about an Isolated Singularity: Let z_0 be an isolated singularity of f. Suppose that f be analytic on the annulus $\mathscr{A} : 0 < r < |z - z_0| < R$. The Laurent's series of f about z_0 is given by

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z-z_0)^n = \underbrace{\sum_{n=1}^{\infty} a_{-n}(z-z_0)^{-n}}_{\text{Principal Part}} + \underbrace{\sum_{n=0}^{\infty} a_n(z-z_0)^n}_{\text{Analytic Part}}, \tag{3.6}$$

where a_n's are called its *coefficients*.

Definition 3.2 (Residue at an Isolated Singularity): The coefficient a_{-1} of the first negative power $1/(z - z_0)$ in (3.9) is known as the *residue* Res $(f; z_0)$ of f at $z = z_0$.

Remark 3.1: The residue should be computed at an isolated singularity, which lies within the region of validity of the Laurent's series. The residue is not defined at an isolated singularity not lying in the validity region.

Laurent's Series of Rational Functions

Remark 3.2 (Isolated singularities of Rational Functions): Consider a rational function $f(z) = \phi(z)/\psi(z)$. If $\psi(z_0) = 0$ and $\phi(z_0) \neq 0$, then z_0 is an isolated singularity of f.

Example 3.5:

(a) Let $f(z) = 1/z$ for all $z \in \mathbb{C}$. Then f is analytic at all points of the complex plane except at $z = 0$, and hence $z = 0$ is an singularity of f.

(b) The function $f(z) = (2z + 3)/(z^2 + 1)$ is analytic on $\mathbb{C}$ except at $z = \pm i$, and $z = \pm\, i$ are isolated singularities of f.

(c) The function $f(z) = (z - 1)/(z + 1)(z - 5)^3$ is analytic on $\mathbb{C}$ except at $z = -1$, 5, and hence $z = -1$, 5 are isolated singularities of f.

Example 3.6: Find the Laurent's series of $f(z) = 1/z^2(1-z)$, valid in the regions $\mathscr{R}_1 : 0 < |z| < 1$, $\mathscr{R}_2 : |z| > 1$, $\mathscr{R}_3 : 0 < |z-1| < 1$ and $\mathscr{R}_4 : |z-1| > 1$. Hence find the residues of f at the isolated singularities.

Solution: Note that $z = 0$ and $z = 1$ are isolated singularities of f.

(a) On $\mathscr{R}_1 : 0 < |z| < 1$:

$$\frac{1}{z^2(1-z)} = \frac{1}{z^2}\sum_{n=0}^{\infty} z^n = \sum_{n=0}^{\infty} z^{n-2} = \frac{1}{z^2} + \frac{1}{z} + 1 + z + z^2 + \cdots .$$

Since $z = 0$ lies within $\mathscr{R}_1$, we find that Res $(f; 0) = a_1 = 1$.

(b) On $\mathscr{R}_2 : |z| > 1$: $\left|\frac{1}{z}\right| < 1$ so that

$$\begin{aligned}\frac{1}{z^2(1-z)} &= -\frac{1}{z^3}\left(1-\frac{1}{z}\right)^{-1} = -\frac{1}{z^3}\left\{1 + \frac{1}{z} + \frac{1}{z^2} + \cdots\right\} \\ &= -\left\{\frac{1}{z^3} + \frac{1}{z^4} + \frac{1}{z^5} + \cdots\right\}.\end{aligned}$$

(c) To find the Laurent's series on $\mathscr{R}_3 : 0 < |z-1| < 1$, write $u = z - 1$ so that $z = u + 1$, and $0 < |u| < 1$. Then

$$\begin{aligned}f(z) &= \frac{1}{(u+1)^2(-u)} = -\frac{1}{u}(1+u)^{-2} = -\frac{1}{u}\sum_{n=0}^{\infty}(n+1)(-1)^n u^n \\ &= -\sum_{n=0}^{\infty}(n+1)(-1)^n u^{n-1} = -\sum_{n=0}^{\infty}(n+1)(-1)^n (z-1)^{n-1}.\end{aligned}$$

Since $z = 1$ lies within $\mathscr{R}_3$, we find that Res $(f; 1) = -1$.

(d) With $u = z - 1$, On $\mathscr{R}_4$, we have $|u| > 1$ or $0 < |1/u| < 1$, and

$$\begin{aligned}f(z) &= \frac{1}{(u+1)^2(-u)} = -\frac{1}{u^3}\left\{1 + \frac{1}{u}\right\}^{-2} = -\frac{1}{u^3}\left\{1 - \frac{2}{u} + \frac{3}{u^2} - \cdots\right\} \\ &= -\frac{1}{u^3} + \frac{2}{u^4} - \frac{3}{u^5} + \cdots = -\frac{1}{(z-1)^3} + \frac{2}{(z-1)^4} - \frac{3}{(z-1)^5} + \cdots\end{aligned}$$

Example 3.7: Find the Laurent's series of $f(z) = 1/(z+1)(z-2)$, valid in the regions $\mathscr{R}_1 : 0 < |z+1| < 3$, $\mathscr{R}_2 : |z+1| > 3$, $\mathscr{R}_3 : 0 < |z-2| < 3$ and $\mathscr{R}_4 : |z-2| > 3$. Hence find the residues of f at the isolated singularities.

Solution: Note that $z = -1$ and $z = 2$ are isolated singularities of f.

(a) Write $u = z + 1$ so that $z = u - 1$. On $\mathscr{R}_1 : 0 < |z+1| < 3$ we have $\left|\frac{u}{3}\right| < 1$. Then

$$\begin{aligned}f(z) &= \frac{1}{u(u-3)} = -\frac{1}{3u}\left\{1 - \frac{u}{3}\right\}^{-1} = -\frac{1}{3u}\left\{1 + \frac{u}{3} + \left(\frac{u}{3}\right)^2 + \cdots\right\} \\ &= -\frac{1}{3}\left\{\frac{1}{u} + \frac{1}{3} + \frac{u}{3^2} + \cdots\right\} = -\frac{1}{3}\left\{\frac{1}{z+1} + \frac{1}{3} + \frac{z+1}{3^2} + \cdots\right\}.\end{aligned}$$

Since $z=-1$ lies within $\mathscr{R}_1$, we find that Res $(f;-1)=a_1=-1/3$.

(b) With $u=z+1$, on $\mathscr{R}_2:|z+1|>3$, we have $\left|\frac{3}{u}\right|<1$, and

$$\begin{aligned} f(z)&=\frac{1}{u(u-3)}=\frac{1}{u^2}\left(1-\frac{3}{u}\right)^{-1}=\frac{1}{u^2}\left\{1+\frac{3}{u}+\frac{3^2}{u^2}+\cdots\right\}\\ &=\frac{1}{u^3}+\frac{3}{u^4}+\frac{3^2}{u^5}+\cdots=\frac{1}{(z+1)^3}+\frac{3}{(z+1)^4}+\frac{3^2}{(z+1)^5}+\cdots. \end{aligned}$$

(c) To find the Laurent's series on $\mathscr{R}_3: 0<|z-2|<3$, write $u=z-2$ so that $z=u+2$ and $0<|u/3|<1$. Therefore,

$$\begin{aligned} f(z)&=\frac{1}{u(u+3)}=\frac{1}{3u}\left(1+\frac{u}{3}\right)^{-1}=\frac{1}{u}\sum_{n=0}^{\infty}(-1)^n(u/3)^n\\ &=\sum_{n=0}^{\infty}(-1)^n(u^{n-1}/3^n)=\sum_{n=0}^{\infty}\left(-\frac{1}{3}\right)^n(z-2)^{n-1}. \end{aligned}$$

Since $z=2$ lies within $\mathscr{R}_3$, we find that Res $(f;2)=1$.

(d) With $u=z-2$, on $\mathscr{R}_4$, we have $|u|>3$ or $0<|3/u|<1$. Therefore,

$$\begin{aligned} f(z)&=\frac{1}{u(u+3)}=\frac{1}{u^2}\left\{1+\frac{3}{u}\right\}^{-1}=\frac{1}{u^3}\left\{1-\frac{3}{u}+\frac{3^2}{u^2}-\cdots\right\}\\ &=\frac{1}{u^3}-\frac{3}{u^4}-\frac{3^2}{u^5}+\cdots=\frac{1}{(z-2)^3}-\frac{3}{(z-2)^4}-\frac{3^2}{(z-2)^5}+\cdots \end{aligned}$$

Laurent's Series about a Point of Analyticity in an Annulus

Example 3.8: Find the Laurent series of $f(z)=1/z(1-z)$ on the annulus $\mathscr{A}:1<|z-2|<2$.

Solution: Note that $z=0$ and $z=1$ are isolated singularities of f which do not belong to the annulus $\mathscr{A}$. We find the Laurent series of f about the point $z=2$ of analyticity as follows:

Write $u=z-2$ or $z=u+2$ and $|u/2|<1$ and $|1/u|<1$ on $\mathscr{A}$. Therefore,

$$\begin{aligned} f(z)&=-\frac{1}{(u+2)(u+1)}=\frac{1}{u+2}-\frac{1}{u+1}\\ &=\frac{1}{2}\left\{1+\frac{u}{2}\right\}^{-1}-\frac{1}{u}\left\{1+\frac{1}{u}\right\}^{-1}\\ &=\frac{1}{2}\sum_{n=0}^{\infty}(-1)^n\left(\frac{u}{2}\right)^n-\frac{1}{u}\sum_{n=0}^{\infty}(-1)^n\left(\frac{1}{u}\right)^n. \end{aligned}$$

The first series converges for $|u/2|<1$ or $|u|<2$, and the second series converges for $|1/u|<1$ or on $|u|>1$. Therefore, the required Laurent's series about $z=2$ on $\mathscr{A}$ is

$$f(z)=\sum_{n=0}^{\infty}\left\{\frac{(-1)^n}{2^{n+1}}\right\}u^n+\sum_{n=0}^{\infty}\frac{(-1)^{n+1}}{u^{n+1}}=\sum_{n=0}^{\infty}\frac{(-1)^n}{2^{n+1}}(z-2)^n+\sum_{n=0}^{\infty}\frac{(-1)^{n+1}}{(z-2)^{n+1}}.$$

3.3 Simply and Multiply Connected Domains

Definition 3.3: A domain $\mathscr{D}$ is said to be *simply connected* if every simple closed contour γ lying entirely can be shrunk to a point without leaving $\mathscr{D}$. A domain which is not simply connected is a *multiply connected* domain.

A simply (doubly, triply, ...) connected region will have no (one, two, ...) holes in it.

Theorem 3.2 (Cauchy's Integral Theorem): *Let $\mathscr{D} \subset \mathbb{C}$ be a simply connected domain. If $f : \mathscr{D} \to \mathbb{C}$ is analytic on $\mathscr{D}$, then $\oint_\gamma f(z)\,\mathrm{d}z = 0$ for any simple closed contour γ in $\mathscr{D}$.*

Theorem 3.3 (Cauchy's Integral Formula): *Suppose that f is analytic in a simply connected domain $\mathscr{D}$ and γ, a simple closed contour lying entirely within $\mathscr{D}$. Then for any point a within γ, we have*

$$\oint_\gamma \frac{f(z)}{z-a}\,\mathrm{d}z = 2\pi i\ f(a) \ \text{ or } \ f(a) = \frac{1}{2\pi i}\oint_\gamma \frac{f(z)}{z-a}\,\mathrm{d}z. \tag{3.7}$$

Theorem 3.4 (Cauchy's Integral Formula for Derivatives): *Under the assumptions of Theorem 3.3, we have*

$$f^{(n)}(a) = \frac{n!}{2\pi i}\oint_\gamma \frac{f(z)}{(z-a)^{n+1}}\,\mathrm{d}z,\ n = 1, 2, 3, \ldots. \tag{3.8}$$

Example 3.9: Compute each of the following integrals around a circle γ:

(a) $\oint_\gamma \frac{\cos z}{z}\,\mathrm{d}z;\ \gamma : |z| = 1,$

(b) $\oint_\gamma \frac{z^2-z+1}{z-1}\,\mathrm{d}z;\ \gamma : |z| = 3/2,$

(c) $\oint_\gamma \frac{\sin z}{(z-\pi/6)^3}\,\mathrm{d}z;\ \gamma : |z| = 1,$

(d) $\oint_\gamma \frac{3z^2+z}{z^2-1}\,\mathrm{d}z;\ \gamma : |z-1| = 1,$

(e) $\oint_\gamma \frac{z-1}{(z+1)^2(z-2)}\,\mathrm{d}z;\ \gamma : |z-i| = 3.$

Solution:

(a) Note that $f(z) = \cos z$ is analytic on $\mathbb{C}$ and $z = 0$ lies within the circle γ. By (3.7),

$$\oint_\gamma \frac{\cos z}{z}\,\mathrm{d}z = 2\pi i \cdot \cos 0 = 2\pi i.$$

(b) Since $z = 1$ lies within γ, by (3.7),

$$\oint_\gamma \frac{z^2-z+1}{z-1}\,\mathrm{d}z = 2\pi i \left| z^2 - z + 1 \right|_{z=1} = 2\pi i \cdot 1 = 2\pi i.$$

(c) Since $z = \pi/6$ lies within γ, by (3.8),

$$\oint_\gamma \frac{\sin z}{(z-\pi/6)^3}\,\mathrm{d}z = \frac{2\pi i}{2!}\cdot\left|\frac{\mathrm{d}^2}{\mathrm{d}z^2}(\sin z)\right|_{z=\pi/6} = \pi i\left|-\sin z\right|_{z=\pi/6} = -\frac{2\pi i}{2} = -\pi i.$$

(d) The point $z = 1$ lies within, whereas $z = -1$ lies outside γ. Therefore, we define $f(z) = (3z^2+z)/(z+1)$ so that

$$\oint_\gamma \frac{3z^2+z}{z^2-1}\,\mathrm{d}z = \oint_\gamma \frac{f(z)}{z-1}\,\mathrm{d}z = 2\pi i\, f(1) = 2\pi i\cdot\frac{3\cdot 1+1}{1+1} = 4\pi i.$$

(e) The points $z = -1$ and $z = 2$ lie inside the circle γ, and by partial fractions,

$$\frac{z-1}{(z+1)^2(z-2)} = -\frac{1}{9}\cdot\frac{1}{z+1} + \frac{2}{3}\cdot\frac{1}{(z+1)^2} + \frac{1}{9}\cdot\frac{1}{z-2}$$

Therefore, by (3.7) and (3.8), we have

$$\begin{aligned}\oint_\gamma \frac{z-1}{(z+1)^2(z-2)}\,\mathrm{d}z &= -\frac{1}{9}\oint_\gamma \frac{1}{z+1}\,\mathrm{d}z + \frac{2}{3}\oint_\gamma \frac{1}{(z+1)^2}\,\mathrm{d}z + \frac{1}{9}\oint_\gamma \frac{1}{z-2}\,\mathrm{d}z \\ &= 2\pi i\left\{-\frac{1}{9}\cdot 1 + \frac{2}{3}\cdot 0 + \frac{1}{9}\cdot 1\right\} = 0\end{aligned}$$

3.4 Residues of a Rational Function at its Poles

Definition 3.4 (Poles): Let $z_0 \in \mathbb{C}$ an isolated singularity[2] of $f(z)$, and

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z-z_0)^n = \underbrace{\sum_{n=1}^{\infty} a_{-n}(z-z_0)^{-n}}_{\text{Principal Part}} + \underbrace{\sum_{n=0}^{\infty} a_n(z-z_0)^n}_{\text{Analytic Part}}. \tag{3.9}$$

be the Laurent's series of f about z_0 on the annulus $\mathscr{A} : 0 < r < |z-z_0| < R$. We say that z_0 is a pole of order m, if the coefficients $a_{-k} = 0$ for $k > m$ and $a_{-m} \neq 0$. A pole of order 1 is a simple pole, poles of order 2 and 3 are double and triple poles respectively.

Theorem 3.5: *An isolated singularity z_0 of a complex function $f(z)$ is a pole of order $m \geq 1$, if and only if $\psi_m(z) = (z-z_0)^m f(z)$ is analytic at z_0, and*

(a) *For $m = 1$,*

$$Res\,(f; z_0) = \lim_{z\to z_0}\psi_1(z) = \lim_{z\to z_0}(z-z_0)f(z), \tag{3.10}$$

(b) *For $m > 1$,*

$$Res\,(f; z_0) = \tfrac{1}{(m-1)!}\Big\{\lim_{z\to z_0}\psi_m^{(m-1)}(z_0)\Big\}. \tag{3.11}$$

[2] $z_0 \in \mathbb{C}$ is an *isolated singularity* of $f(z)$, provided $f(z)$ is analytic on some deleted neighborhood $N(z_0) - \{z_0\}$ of $z_0 \in \mathbb{C}$.

Example 3.10: Find the residues of $f(z) = 1/(z-1)(z^2+1)$ at its poles.

Solution: Note that $z = 1, \pm i$ are zeros of the denominator of order 1, and hence are simple poles of f. Therefore, by (3.10),

(a) $\text{Res}\,(f;1) = \lim_{z\to 1}(z-1)f(z) = \lim_{z\to 1}\left\{\frac{1}{z^2+1}\right\} = \frac{1}{2}$,

(b) $\text{Res}\,(f;i) = \lim_{z\to i}(z-i)f(z) = \lim_{z\to i}\left\{\frac{1}{(z-1)(z+i)}\right\} = \frac{1}{(i-1)(2i)} = \frac{1-i}{4}$,

(c) $\text{Res}\,(f;-i) = \lim_{z\to -i}(z+i)f(z) = \lim_{z\to -i}\left\{\frac{1}{(z-1)(z-i)}\right\} = \frac{1}{(-i-1)(-2i)} = \frac{i+1}{4}$.

Example 3.11: Find the residues of $f(z) = 1/(z+z^2)$ at its poles.

Solution: Note that $z = 0$ and -1 are simple poles of f. Therefore, by (3.10),

(a) $\text{Res}\,(f;0) = \lim_{z\to 0}(z-0)f(z) = \lim_{z\to 0}\frac{1}{z+1} = 1$,

(b) $\text{Res}\,(f;-1) = \lim_{z\to -1}(z+1)f(z) = \lim_{z\to -1}\frac{1}{z} = -1$.

Example 3.12: Find the residues of $f(z) = e^z/(z^2+\pi^2)$ at its poles.

Solution: Note that $z = \pm\pi i$ are simple poles of f. Therefore, by (3.10),

(a) $\text{Res}\,(f;\pi i) = \lim_{z\to \pi i}(z-\pi i)f(z) = \lim_{z\to \pi i}\left(\frac{e^{\pi i}}{z+\pi i}\right) = \frac{i}{2\pi}$,

(b) $\text{Res}\,(f;-\pi i) = \lim_{z\to -\pi i}(z+\pi i)f(z) = \lim_{z\to -\pi i}\left(\frac{e^{-\pi i}}{z-\pi i}\right) = -\frac{i}{2\pi}$.

Example 3.13: Find the residue of $f(z) = e^{2z}/(z-1)^2$ at its pole.

Solution: Since $z = 1$ is a double pole of f, write $\psi_2(z) = (z-1)^2 f(z) = e^{2z}$, by (3.11),

$$\text{Res}\,(f;z_1) = \frac{1}{(2-1)!}\left\{\psi_2'(1)\right\} = \left|2e^{2z}\right|_{z=1} = 2e^2.$$

Example 3.14: Find the residues of $f(z) = 1/(z-1)(z+3)^2$ at its poles.

Solution: Note that $z = 1$ is a simple pole and $z = -3$ is a double pole for f. Therefore, by (3.10),

$$\text{Res}\,(f;1) = \lim_{z\to 1}(z-1)f(z) = \lim_{z\to 1}\frac{1}{(z+3)^2} = 1/16,$$

and

$$\psi_2(z) = (z+3)^2 f(z) = 1/(z-1).$$

By (3.11), we have

$$\begin{aligned}\text{Res}\,(f;-3) &= \frac{1}{(2-1)!}\left\{\psi_2'(1)\right\} = \left[\frac{\mathrm{d}}{\mathrm{d}z}\left\{\frac{1}{z-1}\right\}\right]_{z=-3} \\ &= -\left\{\frac{1}{(z-1)^2}\right\}_{z=-3} = -\frac{1}{16}.\end{aligned}$$

3.5 Cauchy's Residue Theorem

Theorem 3.6 (Cauchy's Residue Theorem): *If f is analytic on a simple (positively oriented) closed contour γ and everywhere inside γ except a finite number of singularities $z_1, z_2, \cdots z_n$, then*

$$\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i \sum_{k=1}^{n} Res\,(f; z_k). \tag{3.12}$$

Example 3.15: Evaluate $\oint_\gamma \frac{1}{(z-1)(z+2)^2}\,\mathrm{d}z$, where γ is the circle

(a) $|z| = 1/2$;

(b) $|z| = 3/2$;

(c) $|z| = 5/2$.

Solution: Define $f(z) = 1/(z-1)(z+2)^2$.

(a) We note that

- f is analytic at all points of the z-plane except at $z = 1, -2$, which are its simple poles.
- The radius of the circle $\gamma : |z| = 1/2$ is $r = 1/2$.
- Since both the poles are at a distance of more than $r = 1/2$ from the centre $z = 0$ of the circle, they lie outside the circle.
- That is, f is analytic within and on the circle γ.
- By Theorem 3.6, $\oint_\gamma \frac{1}{(z-1)(z+2)^2}\,\mathrm{d}z = 0$.

(b) The radius of the circle $\gamma :\ |z| = 3/2$ is $r = 3/2$.

- The pole $z = 1$ lies inside, whereas the pole $z = -2$ lies outside γ.
- By Theorem 3.6, $\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i\,\mathrm{Res}\,(f; 1)$.
- Now $\mathrm{Res}\,(f; 1) = \lim_{z\to 1}(z-1)f(z) = \lim_{z\to 1}\frac{1}{(z+2)^2} = \frac{1}{9}$.
- Hence, $\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i \cdot \frac{1}{9} = \frac{2\pi i}{9}$.

(c) The radius of the circle $\gamma : |z| = 5/2$ is $r = 5/2$

- Both the poles $z = 1, -2$ lie inside γ.
- Now $z = 1$ is a simple pole and $\mathrm{Res}\,(f; 1) = 1/9$.
- whereas -2 is a double pole and $\psi_2(z) = (z+2)^2 f(z) = 1/(z-1)$.

$$\mathrm{Res}\,(f; 1) = \frac{\psi_2'(-2)}{(2-1)!} = \left|\frac{\mathrm{d}}{\mathrm{d}z}\left\{\frac{1}{z-1}\right\}\right|_{z=-2} = \left|-\frac{1}{(z-1)^2}\right|_{z=-2} = -\frac{1}{9}.$$

- Hence, by Theorem 3.6, $\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i[\text{Res}\,(f;1) + \text{Res}\,(f;-2)] = 0.$

Example 3.16: Evaluate $\oint_{|z|=1} \frac{e^z}{\cos\pi z}\,\mathrm{d}z$

Solution: Define $f(z) = e^z/\cos\pi z$. Solving the equation $\cos\pi z = 0$, we get the poles $\pi z = \pm\frac{\pi}{2}$, $\pm\frac{3\pi}{2}, \cdots$, of which $z = \pm 1/2$ lie inside $\gamma : |z| = 1$.

- Now,

$$\text{Res}\left(f;\frac{1}{2}\right) = \lim_{z\to 1/2}(z-1/2)\,f(z) = \lim_{z\to 1/2}\frac{(z-1/2)e^z}{\cos(\pi z)} = -\frac{e^{1/2}}{\pi},$$
$$\text{Res}\left(f;-\frac{1}{2}\right) = \lim_{z\to 1/2}(z+1/2)\,f(z) = \lim_{z\to 1/2}\frac{(z+1/2)\,e^z}{\cos(\pi z)} = \frac{e^{-1/2}}{\pi}.$$

- Hence, by Theorem 3.6,

$$\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i\left[\text{Res}\left(f;\frac{1}{2}\right) + \text{Res}\left(f;-\frac{1}{2}\right)\right] = 2\pi i\left[-\frac{e^{1/2}}{\pi} + \frac{e^{-1/2}}{\pi}\right] = 4i\sinh\frac{1}{2}.$$

Example 3.17: Compute $\oint \frac{3z^3}{(z-1)(z^2+9)}\,\mathrm{d}z$ around the circles $\gamma_1 : |z-2| = 2$ and $\gamma_2 : |z| = 4$.

Solution: Write $f(z) = 3z^3/(z-1)(z^2+9)$. The poles $z = 1, \pm 3i$ are simple.

(a) Consider the circle $\gamma_1 : |z-2| = 2$ with centre at 2 and radius 2.

- The pole $z = 1$ lies inside the circle $\gamma_1 : |z-2| = 2$
- $\text{Res}\,(f;1) = \lim_{z\to 1}(z-1)\,f(z) = \lim_{z\to 1}\frac{3z^3+2}{z^2+9} = -\frac{1}{2}$.
- Hence by Theorem 3.6, $\oint_{\gamma_1} f(z)\,\mathrm{d}z = 2\pi i\,\text{Res}\,(f;1) = 2\pi i(1/2) = \pi i.$

(b) In case of the circle $\gamma : |z| = 4$ with centre at $z = 0$ and radius 4, all the three poles $z = 1,\ \pm 3i$ lie inside γ. Hence by Theorem 3.6,

$$\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i\,[\text{Res}\,(f;1) + \text{Res}\,(f;3i) + \text{Res}\,(f;3i)]$$
$$= 2\pi i\left\{\frac{1}{2} + \frac{15+49i}{2} + \frac{15-49i}{2}\right\} = 6\pi i.$$

Example 3.18: Evaluate $\oint \frac{1}{z^3(z+4)}\,\mathrm{d}z$, around the circles $\gamma_1 : |z| = 2$ and $\gamma_2 : |z+2| = 3$.

Solution: Write $f(z) = 1/z^3(z+4)$. The poles of f are $z = 0, -4$.

Consider the circle $\gamma_1 : |z| = 2$ with centre at 0 and radius 2. Note that The pole $z = 0$ lies inside the circle γ_1. To find the residue at $z = 0$, we employ the Laurent's expansion

$$\begin{aligned}\frac{1}{z^3(z+4)} &= \frac{1}{4z^3}\left(1+\frac{z}{4}\right)^{-1} = \frac{1}{4z^3}\left[1-\frac{z}{4}+\frac{z^2}{16}-\frac{z^3}{64}+\frac{z^4}{256}-\cdots\right]\\ &= \frac{1}{4z^3}-\frac{1}{16z^2}+\frac{1}{64z}-\frac{z}{256}+\cdots\end{aligned}$$

so that Res $(f;0) = a_{-1} = 1/64$. By Theorem 3.6,

$$\oint_{\gamma_1} f(z)\,\mathrm{d}z = 2\pi i\,\mathrm{Res}\ (f;0) = 2\pi i(1/64) = \pi i/32.$$

Both the poles $z = 0, -4$ lie inside the circle $\gamma_2 : |z+2| = 3$ with centre at $z = -2$ and radius 3, and Res $(f;-4) = \lim_{z\to-4}(z+4)f(z) = \lim_{z\to-4}\left\{\frac{1}{z^3}\right\} = -1/64$. Therefore, by Theorem 3.6,

$$\oint_{\gamma_2} f(z)\,\mathrm{d}z = 2\pi i\,[\mathrm{Res}\ (f;0)+\mathrm{Res}\ (f;-4)] = 2\pi i\left\{\frac{1}{64}-\frac{1}{64}\right\} = 0.$$

3.6 Evaluation of Real Integrals

Type 1: $\int_0^{2\pi} Q(\cos t, \sin t)\,\mathrm{d}t$: We reduce Type 1 integral to $\oint_\gamma f(z)\,\mathrm{d}z$, where γ is the unit circle $|z| = 1$, and then employ Theorem ??.

General Procedure

- Let $z = e^{it}$ be any point on $\gamma : |z| = 1$ so that $\mathrm{d}z = ie^{it}\,\mathrm{d}t$ or $\mathrm{d}t = \mathrm{d}z/iz$, where $0 \le t \le 2\pi$.
- Using Euler's identity: $e^{it} = \cos t + i\sin t$, we see that $\cos t = \frac{e^{it}+e^{-it}}{2} = \frac{z^2+1}{2z}$ and $\sin t = \frac{e^{it}-e^{-it}}{2i} = \frac{z^2-1}{2zi}$. Therefore,

$$\int_0^{2\pi} Q(\cos t, \sin t)\,\mathrm{d}t = \oint_\gamma \underbrace{\frac{1}{iz}\cdot Q\left(\frac{z^2+1}{2z}, \frac{z^2-1}{2zi}\right)}_{=f(z)}\mathrm{d}z. \tag{3.13}$$

- Find the poles $z_1, z_2, \cdots, z_n$, of $f(z)$ lying within γ and the residues of $f(z)$ at these the poles, and then employ (??).

Example 3.19: Evaluate $\int_0^{2\pi} \frac{\sin t}{5+4\cos t}\,\mathrm{d}t$.

Solution:

$$I = \int_0^{2\pi} \frac{\sin t}{5+4\cos t}\,\mathrm{d}t = \oint_\gamma \frac{1}{iz}\cdot\frac{(z^2-1)/2iz}{5+4(z^2+1)/2z}\,\mathrm{d}z = -\frac{1}{4}\oint_\gamma \underbrace{\frac{z^2-1}{z(z+1/2)(z+2)}}_{=f(z)}\mathrm{d}z$$

Note that $0, -1/2$ and -2 are simple poles of f, of which 0 and $-1/2$ lie inside γ. Therefore,

$$\mathrm{Res}\ (f;-1/2) = \lim_{z\to-1/2}(z+1/2)f(z) = \lim_{z\to-1/2}\frac{z^2-1}{z(z+2)} = 1,$$

$$\text{Res}\,(f;0) = \lim_{z\to 0} zf(z) = \lim_{z\to 0} \tfrac{z^2-1}{(z+1/2)(z+2)} = -1.$$

In view of (??),

$$I = -\frac{1}{4}\cdot 2\pi i\,[\text{Res}\,(f;-1/2) + \text{Res}\,(f;0)] = -\frac{1}{4}\cdot 2\pi i(-1+1) = 0.$$

Example 3.20: Evaluate $\displaystyle\int_0^{2\pi} \frac{\mathrm{d}t}{2+\sin t}$.

Solution: We have

$$I = \int_0^{2\pi} \frac{\mathrm{d}t}{2+\sin t} = \oint_\gamma \frac{1}{iz}\cdot\frac{1}{2+(z^2-1)/2iz}\,\mathrm{d}z = 2\oint_\gamma \underbrace{\frac{1}{z^2+4iz-1}}_{=f(z)}\mathrm{d}z$$

The poles of f are given by $z^2+4iz-1=0$, which are the simple poles $-2i\pm\sqrt{3}$. But $a=-2i+\sqrt{3}$ lies inside γ. Therefore,

$$\text{Res}\,(f;a) = \lim_{z\to a}(z+a)f(z) = \lim_{z\to -2i+\sqrt{3}} \frac{1}{z-(-2i-\sqrt{3})} = \frac{1}{2\sqrt{3}\,i}.$$

Therefore, by Theorem ??,

$$I = 2\cdot 2\pi i\ \text{Res}\,(f;a) = 2\cdot 2\pi i(1/2\sqrt{3}\,i) = 2\pi/\sqrt{3} = 2\sqrt{3}\,\pi/3.$$

Example 3.21: Evaluate $\displaystyle\int_0^{2\pi} \tfrac{\mathrm{d}t}{\sin^2 t+4\cos^2 t} = \int_0^{2\pi} \tfrac{\mathrm{d}t}{1+3\cos^2 t}$.

Solution: Using (??) and (??), we have

$$I = \oint_\gamma \frac{1}{iz}\cdot\frac{1}{\left\{1+3\frac{(z^2+1)^2}{4z^2}\right\}}\,\mathrm{d}z = \frac{4}{i}\oint_\gamma \frac{z}{3z^4+10z^2+3}\,\mathrm{d}z = \frac{4}{i}\oint_\gamma \underbrace{\frac{z}{(3z^2+1)(z^2+3)}}_{=f(z)}\mathrm{d}z.$$

Note that $\pm i/\sqrt{3}$ and $\pm i\sqrt{3}$ are simple poles of $f(z)$; and $\pm i/\sqrt{3}$ lie inside γ. Therefore,

$$\text{Res}\,(f;i/\sqrt{3}) = \lim_{z\to i/\sqrt{3}}(z-i/\sqrt{3})f(z) = \lim_{z\to i/\sqrt{3}} \tfrac{z}{\sqrt{3}(\sqrt{3}z+i)(z^2+3)} = \tfrac{1}{16},$$

$$\text{Res}\,(f;-i/\sqrt{3}) = \lim_{z\to -i/\sqrt{3}}(z+i/\sqrt{3})f(z) = \lim_{z\to -i/\sqrt{3}} \tfrac{z}{\sqrt{3}(\sqrt{3}z-i)(z^2+3)} = \tfrac{1}{16}.$$

Therefore,

$$I = \frac{4}{i}\cdot 2\pi i\,[\text{Res}\,(f;i/\sqrt{3}) + \text{Res}\,(f;-i/\sqrt{3})] = 8\pi\left(\frac{1}{16}+\frac{1}{16}\right) = \pi$$

Example 3.22: Compute $\displaystyle\int_0^{2\pi} \tfrac{\mathrm{d}t}{a+\cos t}$, where $a>1$.

Solution: Now

$$I = \int_0^{2\pi} \frac{\mathrm{d}t}{a+\cos t} = \oint_\gamma \frac{1}{iz} \cdot \frac{1}{a+\frac{z^2+1}{2z}}\,\mathrm{d}z = \frac{2}{i}\oint_\gamma \underbrace{\frac{1}{z^2+2az+1}}_{=f(z)}\,\mathrm{d}z.$$

The poles of f are $\alpha = -a+\sqrt{a^2-1}$ and $\beta = -a-\sqrt{a^2-1}$.

Note that $\sqrt{a^2-1}$ is the geometric mean of $a-1$ and $a+1$, and hence $a-1 < \sqrt{a^2-1} < a+1$ so that $-1 < \sqrt{a^2-1} - a < 1$, that is $\alpha < 1$, which lies inside γ. Therefore,

$$\text{Res}\,(f;\alpha) = \lim_{z\to\alpha}(z-\alpha)f(z) = \lim_{z\to\alpha}\frac{1}{z-\beta} = \frac{1}{\alpha-\beta} = \frac{1}{2\sqrt{a^2-1}}.$$

Hence

$$\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i\;\text{Res}\,(f;\alpha) = 2\pi i \cdot \frac{1}{2\sqrt{a^2-1}} = \frac{\pi i}{\sqrt{a^2-1}}$$

so that

$$I = \frac{2}{i}\cdot\frac{\pi i}{\sqrt{a^2-1}} = \frac{2\pi}{\sqrt{a^2-1}}.$$

Type 2: $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x$: Let $f(x)$ be a rational function such that its complex form $f(z)$ has a finite number of poles above the real axis.

General Procedure

Recall that $I = \int_{-\infty}^{\infty} f(x)\ \mathrm{d}x = \lim_{R\to\infty}\int_{-R}^{R} f(x)\ \mathrm{d}x$, provided that the limit on the right hand side exists. To evaluate I, we choose a closed contour $\gamma = [-R,R]\cup\gamma_R$, where

(a) $[-R,R]$ is the interval of length $2R$ on the real axis, $R >> 0$ and

(b) γ_R is the arc of the semicircle $|z| = R$ from $(R,0)$ to $(-R,0)$, in the anticlockwise direction in the upper half plane. Along γ_R, $z = e^{it}$, $0 \le t \le \pi$.

Then

$$\oint_\gamma f(z)\,\mathrm{d}z = \int_{-R}^{R} f(x)\ \mathrm{d}x + \int_{\gamma_R} f(z)\ \mathrm{d}z. \tag{3.14}$$

We find the singularities z_1, z_2, ..., z_n of f, which lie inside γ (of course above the real line - why?), and hence the residues there at. Then by residue theorem,

$$\oint_\gamma f(z)\ \mathrm{d}z = 2\pi i \cdot \sum_{j=1}^{n} \text{Res}\,(f; z_j). \tag{3.15}$$

Employing the limit as $R \to \infty$ in (3.14), and using (3.15), we find that

$$I = 2\pi i \cdot \sum_{j=1}^{n} \operatorname{Res}(f; z_j),$$

provided $\left| \int_{\gamma_R} f(z)\,\mathrm{d}z \right| \to 0$ as $R \to \infty$.

Example 3.23: Evaluate $\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^2+1}$.

Solution: By routine integration, we have

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^2+1} = 2\int_{0}^{\infty} \frac{\mathrm{d}x}{x^2+1} = 2\left|\tan^{-1} x\right|_{x=0}^{\infty} = 2\tan^{-1}\infty = 2 \cdot \frac{\pi}{2} = \pi. \tag{3.16}$$

On the other hand, write $f(z) = 1/(z^2+1)$. Then $z = \pm i$ are simple poles of f. Choose the following paths:

- $[-R, R]$: a straight line along the real axis from $-R$ to R
- γ_R: the semi-circular arc in the upper half plane from R to $-R$.
- $\gamma = [-R, R] \cup \gamma_R$.

Note that R is sufficiently large. Then the pole $z = i$ lies inside γ, and

$$\int_{-R}^{R} f(x)\,\mathrm{d}x + \int_{\gamma_R} f(z)\,\mathrm{d}z = \oint_{\gamma} f(z)\,\mathrm{d}z = 2\pi i \cdot \operatorname{Res}(f; i), \tag{3.17}$$

Now,

$$\operatorname{Res}(f; i) = \lim_{z \to i} \frac{1}{(z+i)(z-i)} = \frac{1}{i+i} = \frac{1}{2i}. \tag{3.18}$$

Using $|z^2+1| \geq \left| |z|^2 - 1 \right| = R^2 - 1$, where $R > 1$, we have

$$\left| \int_{\gamma_R} f(z)\,\mathrm{d}z \right| = \left| \int_{\gamma_R} \frac{\mathrm{d}z}{z^2+1} \right| \leq \frac{\pi R}{R^2-1} \to 0 \text{ as } R \to \infty.$$

Therefore, proceeding the limit as $R \to \infty$ in (3.17), and using this

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^2+1} = \pi \text{ so that } \int_{0}^{\infty} \frac{\mathrm{d}x}{x^2+1} = \frac{\pi}{2}.$$

Example 3.24: Verify that $\int_{0}^{\infty} \frac{\mathrm{d}x}{(x^2+a^2)^2} = \frac{\pi}{4a^3}$, $a > 0$.

Solution: Write $f(z) = 1/(z^2+a^2)^2$. Note that $z = \pm\, ai$ are double poles of f. Choose the following paths:

- $[-R, R]$: a straight line along the real axis from $-R$ to R

- γ_R: the semi-circular arc in the upper half plane from R to $-R$.
- $\gamma_R = [-R, R] \cup \gamma_R$

Since the pole $z = ai$ lies inside γ_R above the real axis, we have

$$\int_{-R}^{R} f(x)\,\mathrm{d}x + \int_{\gamma_R} f(z)\,\mathrm{d}z = \oint_{\gamma_R} f(z)\,\mathrm{d}z = 2\pi i \cdot B,$$

where

$$\begin{aligned} B = \operatorname{Res}(f; ai) &= \lim_{z\to ai} \frac{\mathrm{d}}{\mathrm{d}z}\left\{(z-ai)^2 \cdot \frac{1}{(z-ai)^2(z+ai)^2}\right\} \\ &= \lim_{z\to ai} \frac{d}{dz}\left\{\frac{1}{(z+ai)^2}\right\} = -\frac{2}{(2ai)^3} = \frac{1}{4a^3 i}. \end{aligned}$$

Thus

$$\int_{-R}^{R} f(x)\,\mathrm{d}x + \int_{\gamma_R} f(z)\,\mathrm{d}z = \frac{2\pi i}{4a^3 i} = \frac{\pi}{2a^3} \text{ or } \int_{-R}^{R} f(x)\,\mathrm{d}x = \frac{\pi}{2a^3} - \int_{\gamma_R} f(z)\,\mathrm{d}z, \tag{3.19}$$

where $R >> 0$. Now,

$$\begin{aligned} \left|\int_{\gamma_R} f(z)\,\mathrm{d}z\right| &= \left|\int_{\gamma_R} \frac{1}{(z^2+a^2)^2}\,\mathrm{d}z\right| \le \int_{\gamma_R} \frac{1}{|z^2+a^2|^2}\,|\mathrm{d}z| \\ &\le \int_{\gamma_R} \frac{1}{(|z^2|^2 - a^2)^2}\,|\mathrm{d}z| \le \frac{1}{(R^2-a^2)^2}\int_{\gamma_R} |dz| \\ &= \frac{\pi R}{(R^2-a^2)^2} \to 0 \text{ as } R \to \infty. \end{aligned}$$

Proceeding the limit as $R \to \infty$ in (3.19), and then using this, we get

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{(x^2+a^2)^2} = \frac{\pi}{2a^3} \text{ so that } \int_{0}^{\infty} \frac{\mathrm{d}x}{(x^2+a^2)^2} = \frac{\pi}{4a^3}.$$

Example 3.25: Verify that $\int_0^\infty \frac{\mathrm{d}x}{x^4+1} = \frac{\pi}{2\sqrt{2}}$.

Solution: Write $f(x) = 1/(x^4+1)$. The poles of f are given by $z^4 + 1 = 0$ or $z = (-1)^{1/4}$, which are the fourth roots of -1, namely

$$z_k = \cos\left\{\frac{(2k-1)\pi}{4}\right\} + i\sin\left\{\frac{(2k-1)\pi}{4}\right\} \text{ for } k = 1, 2, 3, 4.$$

That is,

$$\begin{aligned} z_1 &= \cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right) = \frac{1+i}{\sqrt{2}}, \\ z_2 &= \cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right) = \frac{-1+i}{\sqrt{2}}, \end{aligned}$$

$$z_3 = \cos\left(\frac{5\pi}{4}\right) + i\sin\left(\frac{5\pi}{4}\right) = \frac{-1-i}{\sqrt{2}},$$
$$z_4 = \cos\left(\frac{7\pi}{4}\right) + i\sin\left(\frac{7\pi}{4}\right) = \frac{1-i}{\sqrt{2}}.$$

Choose the following paths:

- $[-R, R]$: a straight line along the real axis from $-R$ to R
- γ_R: the semi-circular arc in the upper half plane from R to $-R$

to trace the closed contour $\gamma_R = [-R, R] \cup \gamma_R$. The poles z_1 and z_2 only lie above the real axis and inside γ_R. Thus

$$\int_{-R}^{R} f(x)\,\mathrm{d}x + \int_{\gamma_R} f(z)\,\mathrm{d}z = \oint_{\gamma_R} f(z)\,\mathrm{d}z = 2\pi i \cdot (B_1 + B_2),$$

where

$$B_1 = \text{Res }(f; z_1) = \lim_{z\to z_1} (z - z_1)\left(\frac{1}{z^4+1}\right) = \lim_{z\to z_1} \frac{1}{4z^3} = \frac{1}{4z_1^3} = \frac{z_1}{4z_1^4} = -\frac{z_1}{4},$$
$$B_2 = \text{Res }(f; z_2) = \lim_{z\to z_2} (z - z_2)\left(\frac{1}{z^4+1}\right) = \lim_{z\to z_2} \frac{1}{4z^3} = \frac{1}{4z_2^3} = \frac{z_2}{4z_2^4} = -\frac{z_2}{4}.$$

Further,

$$B_1 + B_2 = -\frac{z_1+z_2}{4} = -\frac{1}{4}\left(\frac{1+i}{\sqrt{2}} + \frac{-1+i}{\sqrt{2}}\right) = -\frac{i}{2\sqrt{2}}.$$

Hence

$$\int_{-R}^{R} f(x)\,\mathrm{d}x + \int_{\gamma_R} f(z)\,\mathrm{d}z = -\frac{2\pi i\cdot i}{2\sqrt{2}} = \frac{\pi}{\sqrt{2}} \text{ or } \int_{-R}^{R} f(x)\,\mathrm{d}x = \frac{\pi}{\sqrt{2}} - \int_{\gamma_R} f(z)\,\mathrm{d}z. \tag{3.20}$$

Since

$$\left|\int_{\gamma_R} f(z)\,\mathrm{d}z\right| = \left|\int_{\gamma_R} \frac{1}{z^4+1}\,\mathrm{d}z\right| = \frac{\pi R}{R^4-1} \to 0 \text{ as } R\to\infty.$$

Proceeding the limit as $R \to \infty$ in (3.20), and then using this, we get

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{x^4+1} = \frac{\pi}{\sqrt{2}} \text{ so that } \int_{0}^{\infty} \frac{\mathrm{d}x}{x^4+1} = \frac{\pi}{2\sqrt{2}}.$$

Type 3: $\int_{-\infty}^{\infty} Q(x)\cos ax\,\mathrm{d}x$ **or** $\int_{-\infty}^{\infty} Q(x)\sin ax\,\mathrm{d}x$, where $Q(x)$ is a rational function having no poles on the real axis.

Example 3.26: Verify that $\int_0^{\infty} \frac{x\sin mx\,\mathrm{d}x}{x^2+a^2} = \frac{\pi}{2}e^{-am},\ a > 0, m > 0$

Solution: Let $f(z) = ze^{imz}/(z^2+1)$. Then $z = ai$ is a simple pole above the real axis. Suppose that $\gamma = [-R, R] \cup \gamma_R$ as in the previous problems. Then

$$\oint_\gamma f(z)\,\mathrm{d}z = 2\pi i \cdot \text{Res}\,(f; ai) = 2\pi i \lim_{z\to ai} (z-ai)\frac{e^{iz}}{z^2+1} = 2\pi i \cdot \frac{iae^{-am}}{2ia} = \pi i\ e^{-am}$$

or

$$\int_{-R}^{R} f(x)\,\mathrm{d}x + \int_{\gamma_R} f(z)\,\mathrm{d}z = \pi i\ e^{-am}. \tag{3.21}$$

Any point on γ_R is of the form: $z = Re^{i\psi} = R(\cos\psi + i\sin\psi)$ so that

$$e^{iz} = e^{i\,(R(\cos\psi)+i\sin\psi)} = e^{-R\sin\psi}.e^{i\,R\cos\psi} \text{ so that } \left|e^{iz}\right| = e^{-R\sin\psi}$$

and hence

$$\left|\int_{\gamma_R} f(z)\,\mathrm{d}z\right| \leq \frac{\pi Re^{-R\sin\psi}}{R^2-a^2} \to 0 \text{ as } R\to\infty.$$

In the limit as $R\to\infty$, (3.21) becomes

$$\int_{-\infty}^{\infty}\left\{\frac{x(\cos mx + i\sin mx)}{x^2+1}\right\}\mathrm{d}x = \pi i\ e^{-am}.$$

Comparing the imaginary parts on both sides,

$$\int_{-\infty}^{\infty}\left(\frac{x\sin mx}{x^2+1}\right)\mathrm{d}x = \pi\ e^{-am} \text{ so that } \int_0^{\infty}\left(\frac{x\sin mx}{x^2+1}\right)\mathrm{d}x = \frac{\pi e^{-am}}{2}.$$

Example 3.27: Verify that $\int_0^\infty \frac{\cos 3x}{(x^2+1)^2}\,\mathrm{d}x = \frac{\pi}{e^3}$.

Solution: Note that

$$\int_0^\infty \frac{\cos 3x}{(x^2+1)^2}\,\mathrm{d}x = \frac{1}{2}\int_{-\infty}^{\infty}\frac{\cos 3x}{(x^2+1)^2}\,\mathrm{d}x = \frac{1}{2}\text{Re}\left(\int_{-\infty}^{\infty}\frac{e^{i3x}}{(x^2+1)^2}\,\mathrm{d}x\right).$$

First we evaluate $I = \displaystyle\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x$, where $f(z) = e^{i3z}/(z^2+1)^2$. Then f has a double pole at $z = i$ in the upper half plane, and

$$\text{Res}\,(f; i) = \left|\frac{\mathrm{d}}{\mathrm{d}z}\left\{(z-\mathrm{i})^2 f(z)\right\}\right|_{z=i} = .$$

Therefore,

$$\int_{-\infty}^{\infty}\frac{e^{i3x}}{(x^2+1)^2}\,\mathrm{d}x = 2\pi i\,\text{Res}\,(f;i) = 2\pi i\left|\frac{\mathrm{d}}{\mathrm{d}z}\right|_{z=i} = 2\pi i\left|\frac{e^{i3z}}{(z+\mathrm{i})^2}\right|_{z=\mathrm{i}} = \frac{2\pi}{e^3}$$

Thus $\int_0^\infty \frac{\cos 3x}{(x^2+1)^2}\,\mathrm{d}x = \frac{1}{2}\cdot\frac{2\pi}{e^3} = \frac{\pi}{e^3}$

Type 4: Indented Contours

Theorem 3.7: *Suppose that $f(z)$ has a simple pole $z = c$ on the real axis. If γ_ε is the contour in the parametric form $z = c + \varepsilon e^{i\theta}$ for $0 \le \theta \le \pi$, then*

$$\lim_{\varepsilon\to 0}\int_{\gamma_\varepsilon} f(z)\ dz = \pi\cdot\mathrm{Res}(f;c) \tag{3.22}$$

Proof. Since $z = c$ is a simple pole of f, its Laurent's series is given by

$$f(z) = \frac{a_{-1}}{z-c} + g(z),$$

where $g(z)$ gives its analytic part, and a_{-1} is the residue of f at $z = c$. Hence

$$\int_{\gamma_\varepsilon} f(z)\ \mathrm{d}z = a_{-1}\int_{\gamma_\varepsilon}\frac{1}{z-c}\ \mathrm{d}z + \int_{\gamma_\varepsilon} g(z)\ \mathrm{d}z. \tag{3.23}$$

Now, on γ_ε: $z - c = \varepsilon e^{i\theta}$ so that $\mathrm{d}z = \varepsilon \mathrm{i}e^{\mathrm{i}\theta}\mathrm{d}\theta$ and θ varies from 0 to π. Therefore,

$$\int_{\gamma_\varepsilon}\frac{1}{z-c}\ \mathrm{d}z = \int_0^\pi \frac{1}{\varepsilon e^{i\theta}}\cdot\varepsilon \mathrm{i}e^{\mathrm{i}\theta}\mathrm{d}\theta = \mathrm{i}\int_0^\pi \mathrm{d}\theta = \pi\mathrm{i}. \tag{3.24}$$

And $g(z)$ is analytic at c so that it is differentiable and hence continuous at c, and is bounded in a small neighborhood $N_\delta(c) = \{z : |z-c| < \delta\}$ of c. That is there exists an $M > 0$ such that $|g(z)| \le M$ on $N_\delta(c)$. Hence

$$\left|\int_{\gamma_\varepsilon} g(z)\ \mathrm{d}z\right| \le M\int_{\gamma_\varepsilon}\ \mathrm{d}z = M\cdot\pi\varepsilon \to 0 \text{ as } \varepsilon\to 0. \tag{3.25}$$

Finally, employing the limit as $\varepsilon \to 0$ in (3.23), and then using (3.24) and (3.25), we obtain (3.22). ■

Example 3.28: Show that $\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \pi/2$

Solution: Note that

$$\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \frac{1}{2}\cdot\int_{-\infty}^\infty \frac{\sin x}{x}\,\mathrm{d}x = \frac{1}{2}\cdot\mathrm{Im}\left(\int_{-\infty}^\infty \frac{e^{ix}}{x}\,\mathrm{d}x\right).$$

Let $f(z) = e^{iz}/z$. Then f has a simple pole at $z = 0$. Consider the closed contour $C = \gamma_R \cup [-R, -\varepsilon] \cup \gamma_\varepsilon \cup [\varepsilon, R]$, where γ_R is the circle $|z| = R$ with centre at the origin ans radius R, while γ_ε is the circle $|z| = \varepsilon$ with centre at the origin ans radius ε. Therefore,

$$\oint_C f(z)\ \mathrm{d}z = \int_{\gamma_R} f(z)\ \mathrm{d}z + \int_{-R}^{-\varepsilon} f(x)\,\mathrm{d}x + \int_{\gamma_\varepsilon} f(z)\ \mathrm{d}z + \int_\varepsilon^R f(x)\,\mathrm{d}x. \tag{3.26}$$

As $\varepsilon \to 0$, in view of Theorem 3.7, the third integral tends to

$$-\pi\mathrm{i} \cdot \mathrm{Res}(f, 0) = -\pi\mathrm{i} \lim_{z \to 0} z f(z) = -\pi\mathrm{i} \lim_{z \to 0} e^{iz} = -\pi\mathrm{i} \cdot e^0 = -\pi\mathrm{i}. \tag{3.27}$$

Now we estimate $I_R = \displaystyle\int_{\gamma_R} f(z)\, \mathrm{d}z$. Let $z = x + iy$ so that $e^{iz} = e^{i(x+iy)} = e^{ix}e^{-y}$. Since $|e^{ix}| = \sqrt{\cos^2 x + \sin^2 x} = 1$, it follows that $|e^{iz}| = e^{-y}$. Further, on the arc γ_R, from the parametric form of z, we see that

$$z = Re^{i\phi} \text{ so that } dz = iRe^{i\phi}\mathrm{d}\phi \text{ and } |dz| = R\mathrm{d}\phi$$
$$y = R\sin\phi,\ \phi:\ 0 \to \pi,\ |z| = R$$

Therefore,

$$\left| \int_{\gamma_R} f(z)\, \mathrm{d}z \right| \leq \int_{\gamma_R} \frac{|e^{iz}|}{|z|} \,|\, \mathrm{d}z| \leq \int_0^\pi \frac{e^{-y}}{R} R\mathrm{d}\phi = \int_0^\pi e^{-y}\mathrm{d}\phi.$$

But $\sin\phi \geq 2\phi/\pi$ so that $-R\sin\phi \leq -2R\phi/\pi$ and $e^{-R\sin\phi} \leq e^{-2R\phi/\pi}$. With this substitution, the above inequality becomes

$$\begin{aligned} \left| \int_{\gamma_R} f(z)\, \mathrm{d}z \right| &\leq \int_0^\pi e^{-2R\phi/\pi}\mathrm{d}\phi = 2\int_0^{\pi/2} e^{-2R\phi/\pi}\mathrm{d}\phi \\ &= \frac{\pi(1 - e^{-R})}{R} \to 0 \text{ as } R \to \infty. \end{aligned} \tag{3.28}$$

Moreover, since there is no pole lying inside the contour C, by Cauchy's theorem,

$$\oint_C f(z)\, \mathrm{d}z = 0. \tag{3.29}$$

Hence proceeding the limit as $R \to \infty$ and $\varepsilon \to 0$ in (3.26), and then using (3.27), (3.28) and (3.29), we get

$$0 = 0 - \pi i + \int_\infty^0 \frac{e^{ix}}{x}\,\mathrm{d}x + \int_0^\infty \frac{e^{ix}}{x}\,\mathrm{d}x \text{ or } \int_\infty^\infty \frac{e^{ix}}{x}\,\mathrm{d}x = \pi i.$$

Comparing the imaginary parts both sides, we finally get $\displaystyle\int_0^\infty \frac{\sin x}{x}\,\mathrm{d}x = \pi/2.$